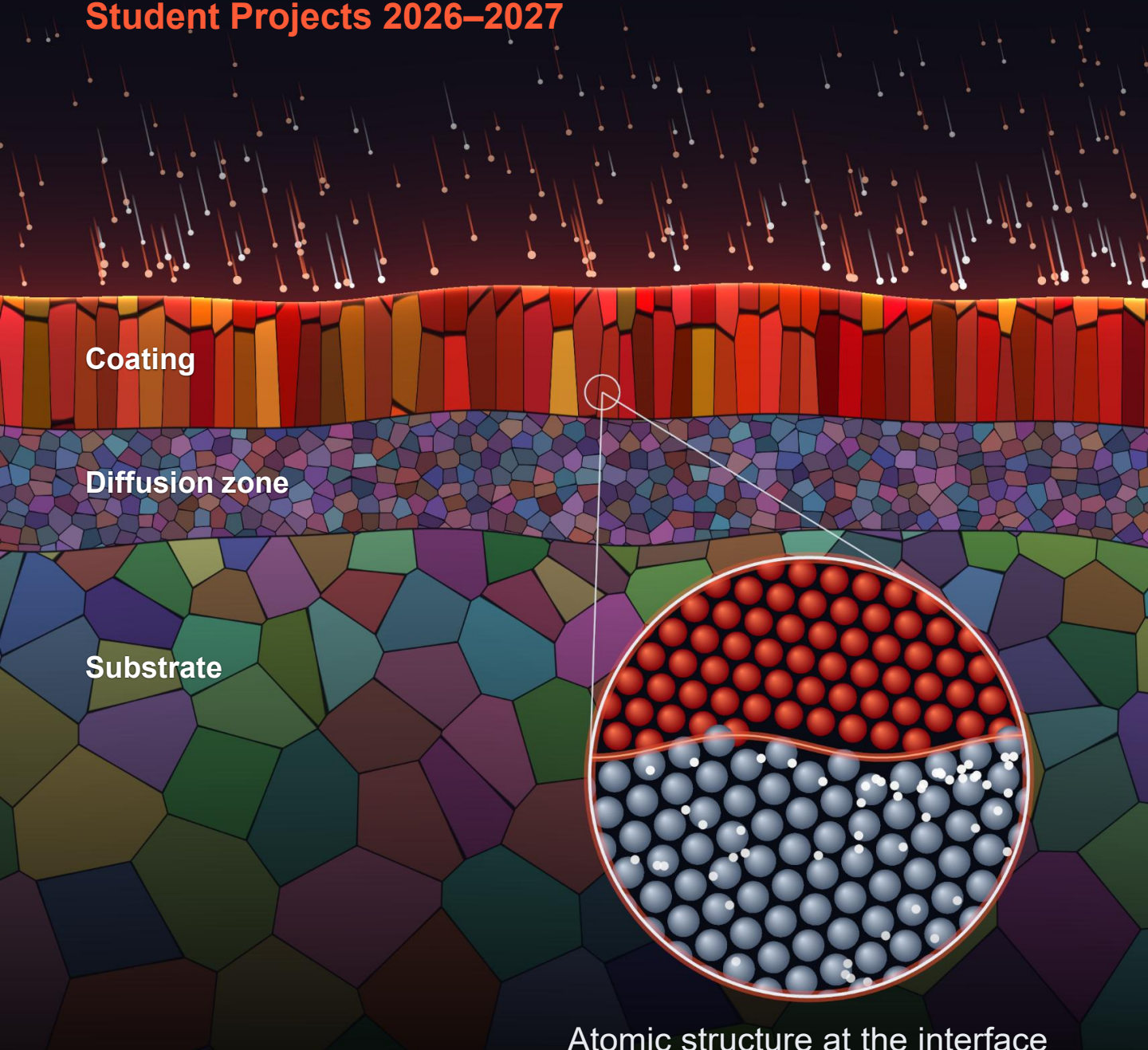
DTU Construct
Department of Civil and Mechanical Engineering

# Materials and Surface Engineering

Materiale- og overfladeteknologi

**Student Projects 2026–2027**




Atomic structure at the interface

The section for Materials and Surface Engineering has expertise in many aspects of materials science and engineering and covers disciplines as materials synthesis, materials solutions, materials characterization, and materials performance. The section teaches and researches in topics spanning from fundamental materials science to innovative materials engineering solutions. The activities include advanced experimental synthesis and characterization techniques as well as modelling and simulation to predict material behavior and design new solutions.

## THE MAIN OVERALL THEMES ARE:

### Surface engineering and materials design

Surface engineering by electrochemical, thermochemical and PVD techniques are applied and further developed. The research activities in surface engineering cover the entire chain from basic research of generic importance to applied research, engineering and innovation.

### Microstructure evolution and phase transformations

Materials microstructure evolution during processing by e.g. thermal treatments or plastic deformation as well as during application lies at the very essence of understanding the relations between process parameters, properties and performance. Understanding the microstructure evolution and phase transformations in materials forms the basis for materials modelling activities.

### Materials performance and degradation

Research in materials performance and materials degradation is focused on corrosion, tribology and high temperature applications, i.e. the chemical, mechanical and thermal interaction of materials with the environment. The degradation mechanisms are investigated and the obtained understanding is explored to predict the life expectancy and design improved materials solutions.

*Currently, we focus on materials for (sustainable) energy applications and materials in medical applications. We have a broad international network of collaboration within academia and industry.*

## PERMANENT SCIENTIFIC STAFF:

**Grethe Winther**
*head of the section*
grwi@dtu.dk

**Rajan Ambat**
ram@dtu.dk

**John Hald**
jhald@dtu.dk

**Morten Stendahl Jellesen**
msje@dtu.dk

**Mehdi Mehrali**
meme@dtu.dk

**Oleg Mishin**
olmi@dtu.dk

**Karen Pantleon**
kapa@dtu.dk

**Wolfgang Pantleon**
pawo@dtu.dk

**Marcel A.J. Somers**
majs@dtu.dk

**Kapil Kumar Gupta**
kkgup@dtu.dk

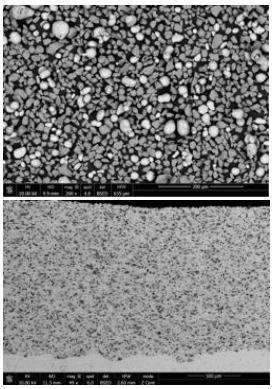
**Cecilie Vase Funch**
cevfu@dtu.dk

**Masoud Hasany**
mahas@dtu.dk

**PROJECT LEVEL: MSc project, BSc project or special course**

# From Metal Powder to Electrical Grids

## Developing functional cold-spray coatings and quantitative methods for powder blends










**SUPERVISORS**

Karen Pantleon
kapa@dtu.dk

Louis Cavichiolo
LC@relibond.com

**COLLABORATION**

Company ReliBond ApS

### PROJECT BACKGROUND

Green energy transition depends on expanding high-capacity electricity grids, including underground and submarine cables connecting offshore renewables to land. Larger conductors enable greater power transmission, but joining them reliably in the field presents a critical installation challenge.
**ReliBond's AlCuCon** connector technology provides fast, reproducible connections across conductor materials and designs. It also supports the transition from copper to more abundant aluminum. Functional cold-spray coatings are central to this technology, and their performance starts with the powder.
**ReliBond** has developed a portable cold-spray platform and established most process parameters.

### PROJECT AIM

Developing functional **cold-spray coatings** and quantitative methods for powder blends:
You will refine Al-, Cu- and Ag-based powder blends with selected filler materials (e.g., $Al_2O_3$), linking composition and homogeneity to coating quality, storage stability and controllable powder flow.
Shape the experimental strategy with DTU and ReliBond.

### PROJECT ACTIVITIES (details to be adapted to the type of project and students' background)

- **Design and test blends:**
  Select fillers and composition ranges using literature and experiments.
  Produce coatings and characterise their quality and functional properties.
- **Develop quantitative methods:**
  Explore methods of materials characterization )e.g. XRD, XRF, ICP-OES, gasometric analysis, ...).
  Develop sample preparation, calibration and measurement procedures.
- **Validate accuracy and mixing endpoints:**
  Use reference blends and representative sampling to quantify uncertainty, distinguish blend variability and determine sufficient mixing times.
- **Connect formulation to performance:**
  Recommend promising blends and preparation procedures, considering segregation, storage, flowability and equipment costs for industrial transfer.

Work alongside ReliBond's team, with access to an office workspace and relevant experimental equipment.

### PRACTICAL NOTES

The overall idea can be subdivided into several projects and, concerning content and time frame, be adapted to the students' education and schedule.

**PROJECT LEVEL: MSc project or special course**

# From waste to value: Electrochemical metal recovery



**SUPERVISORS**

Karen Pantleon
kapa@dtu.dk

Mattia Saccoccio
ms@nitrovolt.com

**COLLABORATION**

NitroVolt ApS, Søborg

## PROJECT BACKGROUND

**NitroVolt** is an electrochemical company, spun out of DTU, that develops and scales electrochemical processes for industrial applications. As part of its continued growth, it is broadening its technology platform into new application areas, including the recovery of metals from industrial waste and side-streams whose metal content is currently unrecovered and unvalued. By extracting these metals, the process keeps valuable resources in the loop, supporting a more circular economy.

The collaboration combines DTU's expertise in electrochemistry, materials characterization and process modelling with NitroVolt's industrial electrochemistry lab and hands-on scale-up experience, giving the student exposure to both academic rigour and industrial relevance.

## PROJECT AIM

The student(s) will contribute to the development of electrochemical metal recovery processes, working on a well-defined sub-problem that fits students' interests and background. The exact scope is defined together with the student and the supervisors at project start, and can range from fully experimental to fully computational, or a combination of both.

## PROJECT ACTIVITIES (details to be adapted to the type of project and students' background)

The project can be formulated around one or a combination of the following topics:

- **Feedstock characterization –** sampling and characterizing industrial side-streams (composition, metal content, impurities) to assess their suitability for electrochemical and/or hydrometallurgical metal recovery.
- **Process design and optimization –** designing, running and optimizing electrochemical recovery experiments (e.g. electrodeposition, electrooxidation/electroreduction) at bench or beaker scale, including electrode, electrolyte and operating-window selection.
- **Physics-informed digital twin modelling –** developing or validating process models (e.g. 1D electrochemical bath models) against experimental data, to support process understanding, design and scale-up decisions.
- **Combined experimental/modelling work –** linking laboratory data to model development or validation, building methodologies that can be reused across chemistries and waste streams.

The project can be based mostly at DTU, mostly at NitroVolt, or split between the two, depending on the chosen direction and the facilities required.

## PRACTICAL NOTES

We are looking for curious and ambitious students who want to develop sustainable solutions and are motivated to work at the interface between experimental science and industrial process development. A background or interest in chemistry, materials science, chemical/mechanical engineering, electrochemistry or process modelling is welcomed.

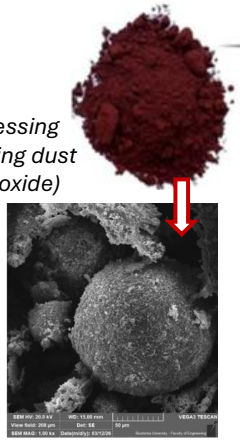
**PROJECT LEVEL: MSc project, BSc project, BEng or special course**

# Sustainable electrodes for hydrogen evolution:
## from biomass and steel industry waste to hydrogen production




*processing of bamboo to synthesize porous carbonaceous material*

*processing pickling dust (iron oxide)*



**SUPERVISOR**

Karen Pantleon

kapa@dtu.dk

**COLLABORATION**

Makerere University, Uganda

**PROJECT BACKGROUND**

The development of **sustainable electrodes**, e.g. for hydrogen production, relies on the use of abundant, naturally occurring raw materials or waste currently disposed of by landfilling, which (until now) are unused resources.
**Bamboo**, a climate resilient and rapidly growing, renewable, natural material, is composed of cellulose, hemicellulose and lignin, which can be decomposed by thermal treatments to form porous carbonaceous materials.
**Pickling dust**, a residue from steel processing industry mainly consisting of iron oxides, can be processed towards the transformation from hematite ($Fe_2O_3$) to magnetite ($Fe_3O_4$) and functionalization to enhance the catalytic activity.
Bamboo and pickling dust, collected in Uganda, are low-cost resources for producing nano-composite electrodes.

**PROJECT AIM** (or smaller steps towards this final goal)

Prototyping sustainable electrodes based on bamboo-derived carbon and pickling waste by testing various production methods, with focus on later upscaling to industrial production in Uganda.

**PROJECT ACTIVITIES**

- **Investigating and optimizing bamboo-derived carbon:**
  Phase identification and chemical analysis of raw and thermally treated bamboo. In particular, the presence of naturally occurring unorganic phases in raw bamboo and their influence on (electrical, electrochemical) properties and morphology of the processed bamboo-derived carbon are of interest.
- **Measurement of relevant properties:**
  For both parts of the final composite electrode, electrical conductivity and active surface area are highly relevant.
- **Optimizing processing routes:**
  Revealing the relationship between processing parameters (e.g. temperature, time, atmosphere) for the thermal treatment of both bamboo (carbonization, activation, doping) and pickling dust (phase transformation, functionalizing). Testing of combined treatments.
- **Testing and comparing** different methods of combining carbon and iron oxide to produce composites with propeties that are relevant for electrodes.

These are suggestions acting as examples for meaningful project(s) or smaller tasks, which will be finally formulated based on the students' background and the time frame of the project. Materials characterization will be essential.

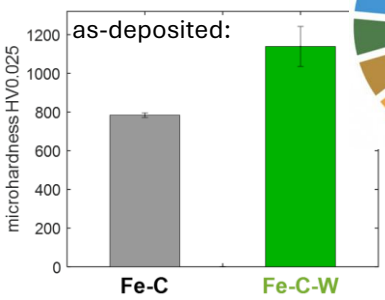
**PRACTICAL NOTES**

The project is carried out at DTU, and include online collaboration with project partners in Uganda.

**PROJECT LEVEL: MSc project, BSc project or special course**

# Surface engineering with Fe-C-based alloy coatings by sustainable electroplating








**SUPERVISOR**

Karen Pantleon
kapa@dtu.dk

## PROJECT BACKGROUND

For surface protection from mechanical and/or chemical damage, electroplating has a long tradition in industry. Present challenges concern the replacement of traditional but harmful electrolytes to sustainable plating using electrolytes, which are not hazardous to neither human health nor environment – without compromising coating properties.

**Novel electroplated iron-based coatings** (recently developed at DTU Construct) utilize the intentional co-deposition of carbon from citric acid containing electrolytes, which results in high carbon contents of about 0.8 wt% C. The as-deposited Fe-C coatings provide excellent very high microhardness, which further increases by dedicated post-deposition annealing.

## PROJECT AIM

Depending on the students' background and interest, different projects are possible aiming at

EITHER - development of electrolytes incl. testing of plating abilities

OR - materials characterization of as-deposited and post-deposition treated coatings from sustainable plating

## PROJECT ACTIVITIES (examples)

Suggestions for sustainable electroplating of iron-carbon-based coatings:

- **Work in the galvano-lab at DTU:**
  Testing and further developing non-hazardous and environmentally friendly electrolytes.
- **Post-deposition annealing of as-deposited samples:**
  Systematic thermal treatments at either isothermal or isochronal conditions, incl. materials characterization.
- **Complementary materials characterization of electroplated samples:**
  Perform materials characterization based on microscopy, spectroscopy and diffraction analysis and relate the coatings' internal structure to the applied parameters of electroplating.
- **Characterisation of relevant properties:**
  Depending on the type of electroplated materials and the applied substrates, mechanical and/or electrochemical and/or electrical properties would be relevant.

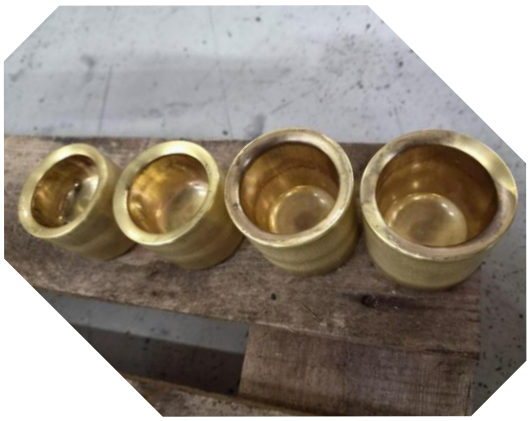
Sustainable plating of coatings other than Fe-C-based alloys ... - let´s discuss.

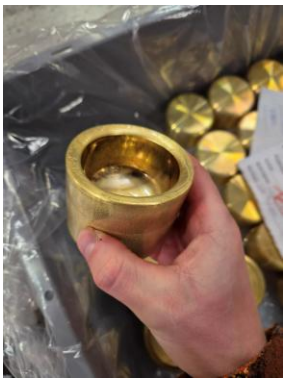
## PRACTICAL NOTES

Several projects are possible with content and time frame, depending on the students' education.

**PROJECT LEVEL: MSc project, BSc project, BEng project or special course**

# Relation between microstructure and hardness for cold-forging of brass components





**SUPERVISORS**

Karen Pantleon
kapa@dtu.dk

Chris Valentin Nielsen
cvni@dtu.dk

Casper Hasle Clausen
ccl@Steelproducts.dk

**COLLABORATION**

Steel Products, Horsens

## PROJECT BACKGROUND

More than 100 years ago, the Danish company Steel Products was established and over the years has specialized in manufacturing products made of different alloys, not only steels. One of its present developments addresses brass components, which rely on the interplay between deformation and annealing. This meets new challenges of understanding and optimizing the relationship between the materials' microstructure – properties – processing – performance of the final brass product.

For cold-forging brass components, their initial microstructure and related properties are crucial.
In general, the expectation is, the softer the material, the easier to cold-forge. However, annealing of the raw material makes some brass alloys harder and this results in a more difficult cold-forging process. Understanding this phenomenon from a material's perspective and suggesting the most suitable raw material characteristics is of interest.

## PROJECT AIM

Relate microstructure and mechanical properties of Cu-Zn alloys (brass) to their ability of cold-forging industrial products.
Suggest the optimal characteristics of the raw material for optimized cold-forging.

## PROJECT ACTIVITIES

With focus on real industrial products, the following topics are suggested:

- **Investigation of as-delivered brass in relation to cold-forging:**
  Delivery conditions of several batches of brass can vary a lot in microstructure and associated hardness, despite the same chemical composition of the Cu-Zn alloy.
  Systematic analysis of as-delivered alloys using microscopy (LOM and SEM) and hardness testing should be carried out and be linked to the cold-forging ability of the alloys. Tensile testing might be an additional option (for longer projects).

- **Site-specific materials characterization of industrial products as a function of processing steps:**
  During cold-forging, the applied brass alloy(s) are subjected to different degrees of deformation, resulting in varying microstructure and properties within a product. Additional changes are expected due to annealing in-between various steps of deformation in the progress of manufacturing.
  Site-specific materials characterization should be linked to the applied processing.

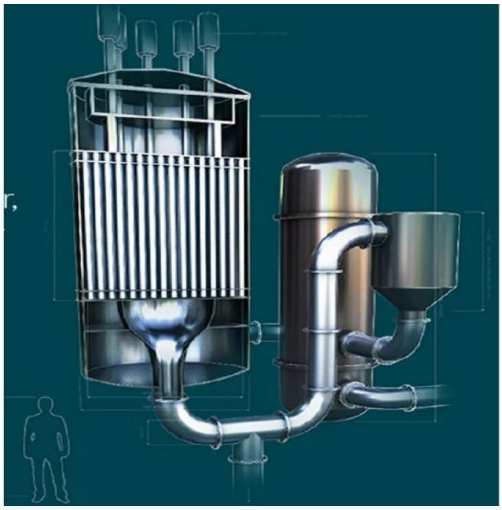
## PRACTICAL NOTES

The work is carried out at DTU, in close collaboration with the company.

PROJECT LEVEL: Master and Bachelor

# Corrosion in Molten Salt







SUPERVISORS

Morten S. Jellesen
msje@dtu.dk,
John Hald
jhald@dtu.dk,
Birgitte Stoffersen
birst@dtu.dk,

COLLABORATION

Molten Salt companies as Copenhagen Atomics, Seaborg Technologies or Nordic Salt Cycle

## 01 / PROJECT BACKGROUND

Molten salt technologies offers new promising qualities, like excellent heat transfer properties or inherent safety arising from high thermal stability and low pressures. The disadvantage of molten salt lies in their high corrosivity, which makes their long-term containment challenging. Nickel alloys and nickel surface treatments are the most promising solutions; however, further development is necessary.

## 02 / PROJECT AIM

[The scope of this master thesis project is to investigate the corrosion resistance in molten salt, e.g.

- Characterize samples in SEM/EDS and optical microscope.
- Conduct corrosion tests in molten fluorides at high temperatures.
- Investigate corrosion resistance in SEM/EDS and optical microscope.
- Make recommendations about critical values and further work.

## 03 / PROJECT ACTIVITIES

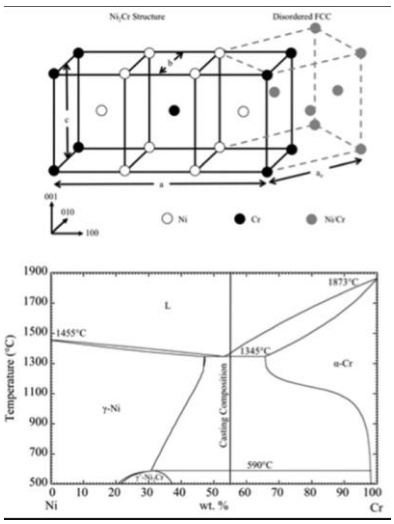
- All the projects involve literature study, experimental exposure tests, preparation of samples for characterization by SEM and XRD, determination of the nature of the corrosion and corrosion rates

## 04 / PRACTICAL NOTES

The exact topic and nature of the project will be specified in a discussion with the students who are interested.

**PROJECT LEVEL: Master and Bachelor**

# Short- and Long-range Ordering of Ni-Cr-Fe Alloys For Nuclear Reactors And Energy Storage Plants




**SUPERVISORS**

Morten S. Jellesen
msje@dtu.dk,
John Hald
jhald@dtu.dk,
Birgitte Stoffersen
birst@dtu.dk,

**COLLABORATION**

Molten Salt companies as Copenhagen Atomics, Seaborg Technologies or Nordic Salt Cycle

## 01 / PROJECT BACKGROUND

The focus of this project is an investigation of the short-range ordering (SRO) and long-range ordering (LRO) that can occur in Ni-Cr-Fe alloys. Industry applications such as next-generation salt-moderated nuclear power plants and salt thermal energy storage systems are highly corrosive environments relying on Ni-Cr-Fe alloys for tubes and tanks.

## 02 / PROJECT AIM

SRO and LRO are phenomena that can occur in Ni-Cr alloys where the nickel and chromium atoms arrange in a non-stoichiometric $Ni_2Cr$ phase. This type of ordering is known to occur after long-time exposure to temperatures between 350-550°C. The ordering can lead to severely decreased mechanical properties by negative creep and promoted planar slip resulting in catastrophic failure of the affected part.

## 03 / PROJECT ACTIVITIES

The project is experimental but will also require the student to apply knowledge from literature to find a viable solution to the challenges. The experimental part could include:

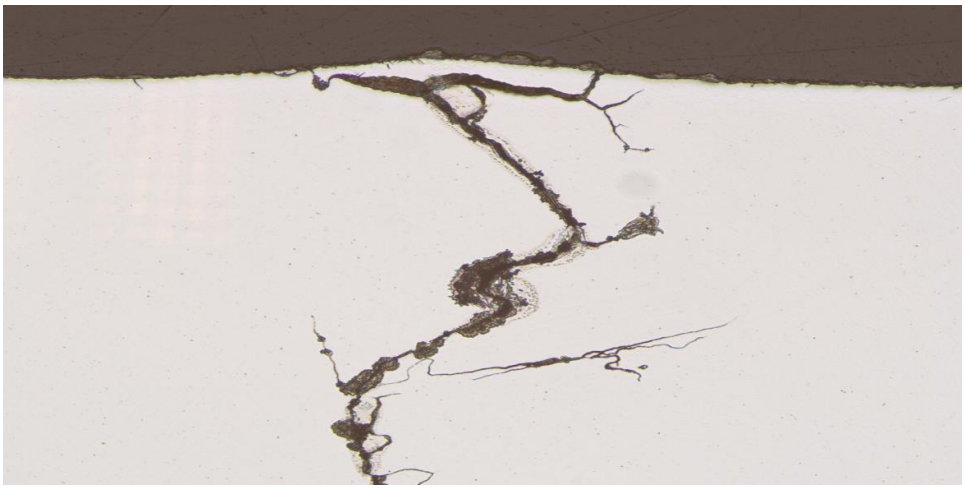
- Heat treatment of relevant Ni-Cr tubes and coupons using furnaces.
- Evaluation of the developed ordering and the severity of the ordering.
- Characterization of the heat-treated alloys using: LOM, XRD, SEM, EDS, TEM and micro-hardness

## 04 / PRACTICAL NOTES

The exact topic and nature of the project will be specified in a discussion with the students who are interested.

**PROJECT LEVEL: Master and Bachelor**

# Stress Corrosion Cracking (SCC) in Welds For Nuclear Reactors



**SUPERVISORS**

Morten S. Jellesen
msje@dtu.dk,
John Hald
jhald@dtu.dk,
Birgitte Stoffersen
birst@dtu.dk,

**COLLABORATION**

Molten Salt companies as Copenhagen Atomics, Seaborg Technologies or Nordic Salt Cycle

## 01 / PROJECT BACKGROUND

The focus of this project is an investigation of stress corrosion cracking (SCC) in nickel welds and in the heat-affected zone (HAZ) around the welds . Industry applications such as next-generation molten salt-moderated nuclear power plants and molten salt thermal energy storage systems are highly corrosive environments relying on Ni-Cr-Fe alloys for tubes and tanks.

## 02 / PROJECT AIM

Welded areas are known weak spots where residual stresses can build up leading to SCC. SCC can occur suddenly and without warning leading to a catastrophic failure. The project is highly experimental but will also require the student to apply known knowledge from literature to find a viable solution to the challenges.

## 03 / PROJECT ACTIVITIES

The experimental part could include:

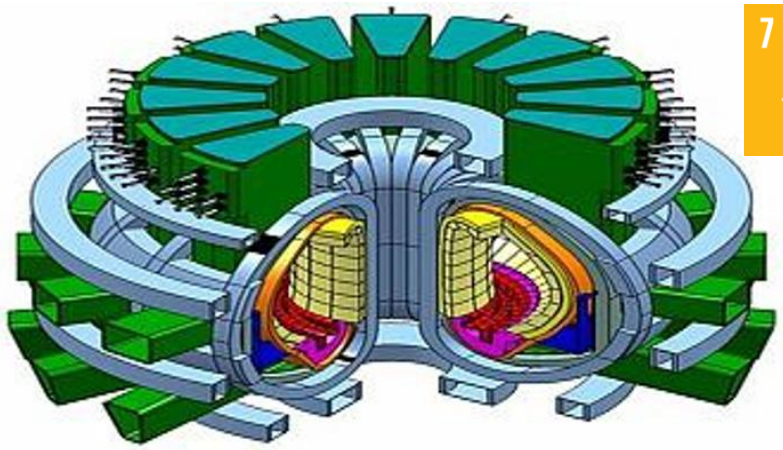
- Corrosion of welded Ni-Cr-Fe tubes and coupons in NaOH.
- Developing a heat treatment for the Ni-welds using both furnaces and heat wrapping.
- Evaluation of the developed heat treatment and on the necessity of a heat treatment.
- Characterization of the welds and the HAZ using; LOM, XRD, SEM, EDS, and micro-hardness.

## 04 / PRACTICAL NOTES

The exact topic and nature of the project will be specified in a discussion with the students who are interested.

**PROJECT LEVEL: Master or Bachelor**

# Advanced materials for fusion reactors



7 AFFORDABLE AND CLEAN ENERGY

**SUPERVISORS**

Wolfgang Pantleon
pawo@dtu.dk

Karen Pantleon
kapa@dtu.dk

**COLLABORATION**

EUROfusion consortium

## 01 / PROJECT BACKGROUND

One of the key issues in constructing fusion reactors are materials able to withstand high temperatures as well as high mechanical and radiation load. The project focusses on the microstructural property relationships of materials relevant for future fusion reactors. Tungsten and tungsten alloys are considered as sole option for the plasma facing armor of the first wall material and the divertor. In service, the materials must withstand high temperatures significantly altering their microstructure.

## 02 / PROJECT AIM

The replacement of the instable microstructure leads to degradation of the mechanical properties as a loss in strength and embrittlement. The aim of the project is to investigate the thermal stability of tungsten, tungsten alloys and tungsten composites processed in different manner (highly deformed by rolling or wire drawing, chemical vapor deposited, powder metallurgical or additively manufactured) with focus on the microstructural mechanisms (recovery, recrystallization and grain growth) relevant for property degradation.

## 03 / PROJECT ACTIVITIES

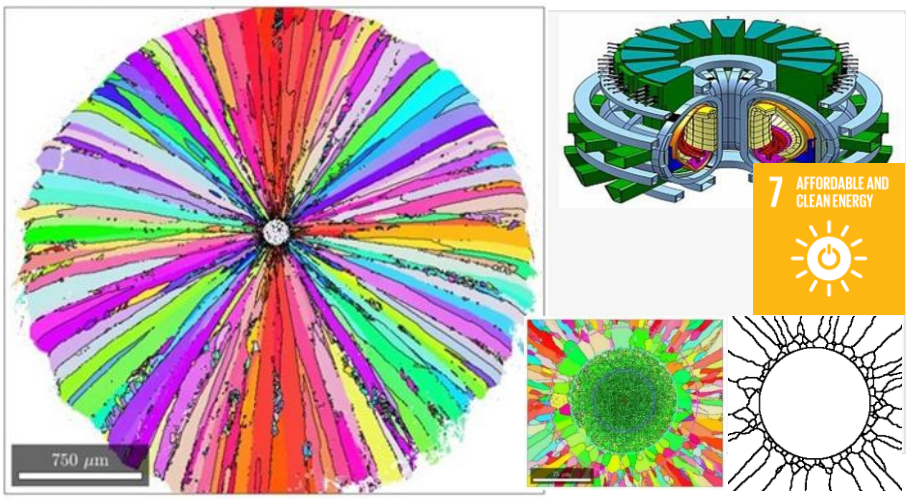
- Rolled tungsten plates or drawn tungsten wires with or without potassium doping
- Dispersion-strengthened tungsten by WC, ZrC, $Y_2O_3$
- Tungsten fiber-reinforced tungsten composites (continuous vs. short fibers, porous vs. compact matrix)

## 04 / PRACTICAL NOTES

The microstructure and mechanical properties of the as-processed material as well as after heat treatment will be analyzed in detail by light optical microscopy, electron backscatter diffraction, and hardness testing. In-situ monitoring the changes at high temperatures by X-ray diffraction will be attempted.

PROJECT LEVEL: Master or Bachelor

# Monte Carlo simulation of grain growth in composites




SUPERVISORS

**Wolfgang Pantleon**
pawo@dtu.dk

**Dana Zöllner**
dana.zoellner@ovgu.de

COLLABORATION

Otto von Guericke University, Magdeburg

## 01 / PROJECT BACKGROUND

Tungsten fiber-reinforced tungsten composites with a tungsten matrix reinforced by heavily drawn tungsten fibers are developed for future fusion reactors. During service at high operation temperatures, the microstructure of the highly deformed fibers will be altered by recrystallization and grain growth. In composites where tungsten was chemically vapor deposited on a single drawn tungsten wire, a puzzling interplay of several physical processes (recrystallization in the micrometer-sized wires and grain growth in the matrix) affects the microstructure at different length scales.

## 02 / PROJECT AIM

Both processes occur in a coupled fashion in the interfacial zone on an even smaller length scale. To address the coupling of multiple metallurgical processes on multiple length scales computational methods are required. The aim of the project is to perform Monte Carlo simulations of the microstructural evolution at high temperatures based on the real microstructure revealed by electron backscatter diffraction and to predict the evolving texture.

## 03 / PROJECT ACTIVITIES

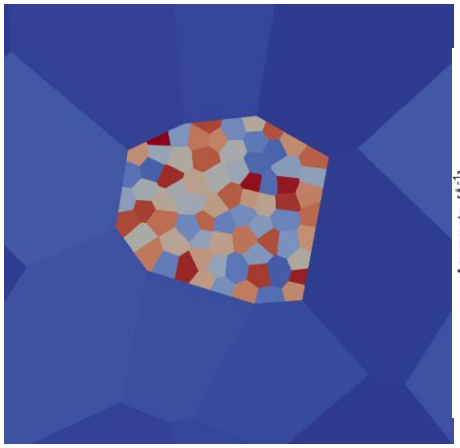
- Setting up a digital twin of the composite for MCS based on electron backscatter data
- Simulating the microstructural evolution in 3D using Potts model by MCS
- Variation of materials parameters to mimic the experimental observations

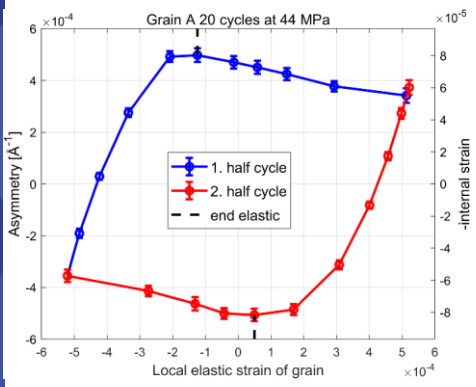
## 04 / PRACTICAL NOTES

By varying grain boundary energy and mobility as well as nucleation rates for recrystallization, the observed peculiar behavior will be rationalized, the importance of the different metallurgical process (primary vs. secondary recrystallization or abnormal grain growth) revealed and the effect of the process geometry assessed.

PROJECT LEVEL: [Master or Bachelor]

# Simulation of internal stresses developing during fatigue






SUPERVISORS

**Wolfgang Pantleon**
pawo@dtu.dk

**Ricardo Lebensohn**
lebenso@lanl.gov

COLLABORATION

Los Alamos National Laboratory, USA

## 01 / PROJECT BACKGROUND

During plastic deformation of metals, the defects carrying plastic deformation are stored in dislocation walls separating dislocation-free subgrains. A peculiar phenomenon has been discovered during fatigue when the sense of the deformation is reversed repeatedly. Even small plastic strains applied repeatedly e.g. in tension compression cycles cause formation of ordered dislocation structures which after a large number of cycles eventually lead to failure.

## 02 / PROJECT AIM

Simultaneously with emerging dislocation structures, internal stresses develop. Monitoring these by in-situ synchrotron diffraction during tension-compression cycles showed their increase during elastic unloading. Neither classical models, nor elasto-viscoplastic simulations have been able to rationalize this unexpected behavior. The reason has been traced to local variations of material parameters and internal stresses. Aim of the project is their incorporation in the simulation and the development of dedicated analysis to allow comparison with experimental X-ray diffraction profiles.

## 03 / PROJECT ACTIVITIES

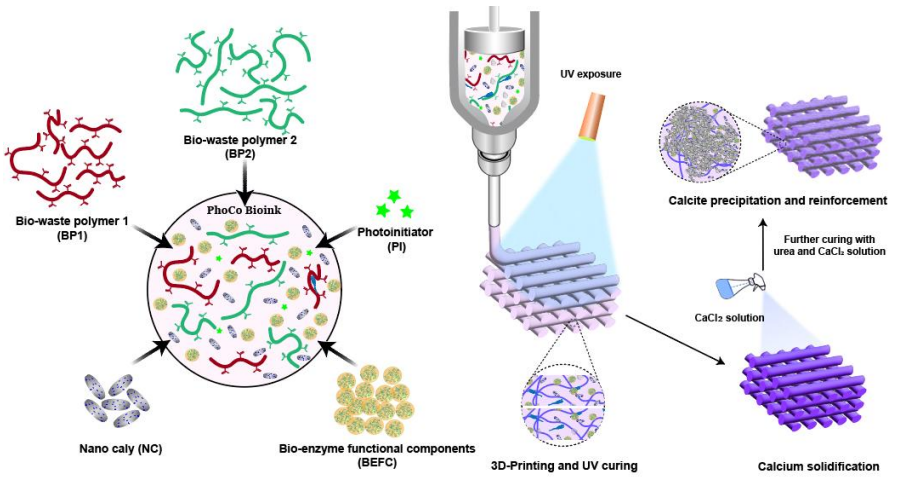
- Large strain elasto-viscoplastic simulations using Fast Fourier transformation approaches
- Varying material properties with focus on intrinsic variability
- Analyzing simulation results for distributions of local elastic strains along loading direction

## 04 / PRACTICAL NOTES

The project is focused on simulations and post-processing of simulation results to derive characteristic profiles which can be compared to already obtained experimental data by in-situ synchrotron diffraction.

PROJECT LEVEL: Master or bachelor

# 3D Printing of Living Materials for Future Sustainable Construction




SUPERVISORS

**Mehdi Mehrali**
meme@dtu.dk

**Peng Jin**
penji@dtu.dk

## 01 / PROJECT BACKGROUND

Conventional building materials, particularly Portland cement, contribute significantly to global carbon dioxide emissions. Developing alternative binders is therefore an important step towards more sustainable construction. Inspired by the formation of natural materials such as bone and nacre, this project explores biological calcium carbonate precipitation within 3D-printed hydrogel structures. Biomass-derived hydrogels provide a printable framework that carries biological components. Under suitable conditions, these components promote calcium carbonate formation within the hydrogel network, potentially producing a denser and stronger composite material.

## 02 / PROJECT AIM

The project aims to develop biomass-derived hydrogels suitable for 3D printing and compatible with biological mineralisation. The student will investigate how hydrogel composition and mineralisation conditions influence printability, calcium carbonate formation, microstructure and mechanical performance, with the broader goal of developing living materials for sustainable construction.

## 03 / PROJECT ACTIVITIES

- Develop and optimise hydrogel formulations for printability, shape stability and compatibility with biological components.
- Fabricate 3D-printed structures and investigate conditions that promote calcium carbonate precipitation within the hydrogel network.
- Monitor mineralisation and evaluate its effects on microstructure, bonding mechanisms and mechanical properties.

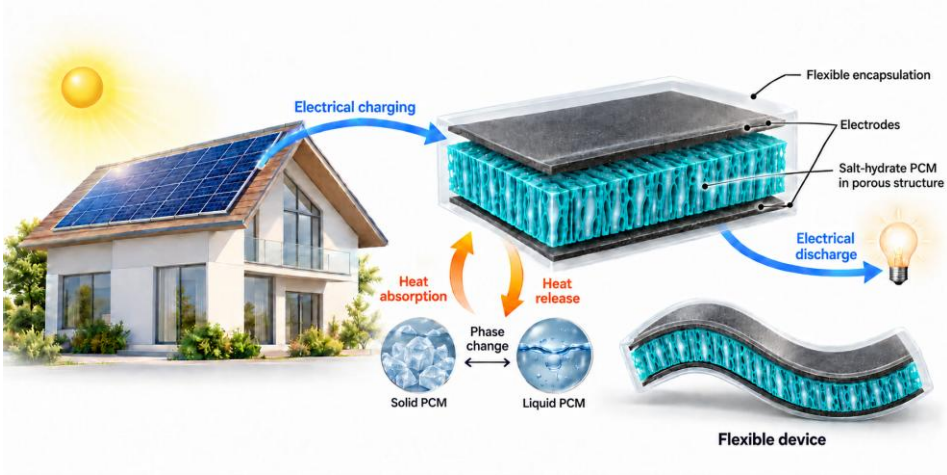
## 04 / PRACTICAL NOTES

The project is primarily experimental and involves hydrogel preparation, 3D printing, controlled mineralisation and materials characterisation. Depending on the research focus, methods may include computed tomography (CT), X-ray diffraction (XRD), scanning electron microscopy (SEM), rheological measurements and mechanical testing.

An interest in chemistry, materials science and sustainable construction is encouraged. Specific activities can be adapted to the student's academic background and interests.

PROJECT LEVEL: Master or bachelor

# Flexible Phase Change Composites for Simultaneous Electrical and Thermal Energy Storage




SUPERVISORS

Mehdi Mehrali
meme@dtu.dk

Amir Reza Akhiani
amrekah@dtu.dk

## 01 / PROJECT BACKGROUND

Phase change materials (PCMs) absorb and release substantial amounts of latent heat at nearly constant temperatures during phase transitions. This makes them attractive for thermal energy storage and temperature regulation in applications such as buildings, solar energy systems, clothing and electronics. Integrating PCMs into flexible energy storage devices offers the opportunity to combine thermal management with electrical energy storage. This project explores salt-hydrate-based composites in which the PCM is intended to function both as a thermal energy reservoir and as an electrolyte within an electrochemical storage device. Incorporating these materials into a flexible foam structure could provide mechanical support and help retain the PCM during operation.

## 02 / PROJECT AIM

The project aims to develop and characterise flexible phase change composites capable of supporting both thermal and electrical energy storage. The student will investigate how salt hydrate composition, foam structure and encapsulation influence thermal storage capacity, electrochemical performance, flexibility and cycling stability.

## 03 / PROJECT ACTIVITIES

- Prepare and optimise flexible PCM composites using different salt hydrates and suitable foam structures.
- Develop encapsulation strategies to retain the PCM, minimise leakage and maintain flexibility during repeated phase transitions.
- Fabricate prototype devices and evaluate their thermal and electrical energy storage performance, including stability during repeated cycling and bending.

## 04 / PRACTICAL NOTES

The project is predominantly experimental and involves composite preparation, PCM incorporation, encapsulation and device testing. Characterisation methods may include differential scanning calorimetry (DSC) to assess phase transitions and latent heat, X-ray diffraction (XRD) to investigate crystalline phases, and scanning electron microscopy (SEM) to examine the composite microstructure. Electrochemical measurements and mechanical testing will help evaluate energy storage performance and flexibility.
This project is suitable for students interested in energy storage and functional materials. The research focus can be tailored to the student's background and learning goals.

PROJECT LEVEL: Master or bachelor

# Smart Energy Building: Design and Fabrication of Quasi-solid-state Supercapacitor




SUPERVISORS

Mehdi Mehrali
meme@dtu.dk

## 01 / PROJECT BACKGROUND

Integrating energy storage into building materials offers opportunities to develop multifunctional construction components. Alongside their conventional roles, these materials could store electrical energy, support the use of renewable energy and provide short-term backup power.
Quasi-solid-state supercapacitors use gel electrolytes that enable ion transport while reducing the risk of electrolyte leakage compared with liquid systems. This project explores electrode and gel electrolyte materials for supercapacitors intended for integration into building components, combining materials development with device fabrication and performance evaluation.

## 02 / PROJECT AIM

The project aims to design, fabricate and characterise a quasi-solid-state supercapacitor for potential integration into building materials. The student will investigate how electrode composition, gel electrolyte properties and their interfaces influence capacitance, charge–discharge behaviour and cycling stability.

## 03 / PROJECT ACTIVITIES

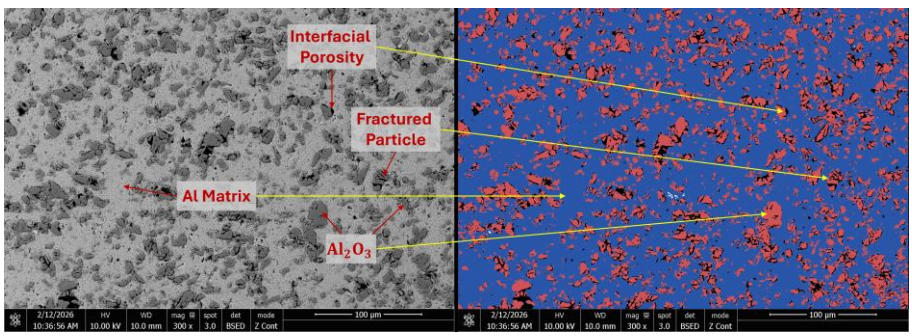
- Prepare and optimise electrode materials and gel electrolytes suitable for quasi-solid-state supercapacitors.
- Characterise the physicochemical properties of the materials and investigate electrode–electrolyte compatibility.
- Assemble prototype devices and evaluate their electrochemical performance, including capacitance, rate capability and cycling stability.

## 04 / PRACTICAL NOTES

The project is primarily experimental and involves materials preparation, device assembly and physicochemical and electrochemical characterisation. Depending on the research focus, methods may include Fourier-transform infrared spectroscopy (FTIR), contact angle measurements, X-ray photoelectron spectroscopy (XPS) and X-ray diffraction (XRD). Electrochemical testing may include cyclic voltammetry, galvanostatic charge–discharge measurements and electrochemical impedance spectroscopy.This project is suitable for students interested in energy storage and functional materials. The research focus can be tailored to the student's background and learning goals.
Students will gain hands-on experience in materials preparation, supercapacitor fabrication and electrochemical testing. Specific activities can be adjusted to match the student's interests and experience.

PROJECT LEVEL: MASTER'S THESIS

# Powder design for cold-sprayed Al-$Al_2O_3$ coatings on aluminum substrate




Left: BSE cross-section of a coating deposited from an Al-60 wt.% $Al_2O_3$ powder blend. Right: Corresponding image segmentation used to quantify the area fractions of the Al matrix, retained $Al_2O_3$ and porosity.

SUPERVISORS

**Lead Supervisor**

**Marcel A.J. Somers**
DTU Construct
majs@dtu.dk

**Co-supervisors**

**Basit Ali**
DTU Construct
baali@dtu.dk

**Louis Cavichiolo**
ReliBond

COLLABORATION

ReliBond ApS, Denmark

## 01 / PROJECT BACKGROUND

High-voltage cables transmit electricity in underground power networks, subsea links and connections to offshore wind farms. Since cables are manufactured and installed in sections, their conductors must be joined to maintain reliable electrical and mechanical continuity. Welding can create these connections, but its heat input can damage nearby insulation and complicate on-site installation. Cold spraying offers a way to form metallic contact layers without melting the powder. Relibond ApS, Denmark develops cold-sprayed aluminum coatings for high-voltage cable joints. Alumina in the feedstock assists surface activation, but excessive retention and unstable powder delivery can limit coating quality .

## 02 / PROJECT AIM

Establish how alumina content in the powder feedstock influences coating formation, microstructure and adhesion on aluminum conductors. Identify a suitable blend that combines reliable feeding and stable nozzle operation with strong bonding and low retained alumina. Support this recommendation through validated measurements of blend composition and homogeneity, together with repeat coating experiments.

## 03 / PROJECT ACTIVITIES

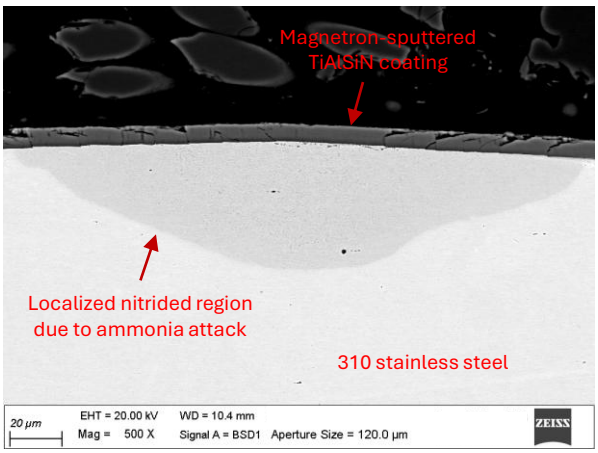
- Prepare Al-$Al_2O_3$ powder blends with varying alumina contents, with limited particle-size comparisons where feasible.
- Develop and validate a method to verify blend homogeneity and the powder composition delivered by the feeder.
- Produce coatings under controlled cold-spray conditions and evaluate deposition efficiency, coating build-up and nozzle operability.
- Use SEM/EDS, image analysis and pull-off testing to assess retained alumina, porosity and adhesion. Compare promising blends with the reference through repeat experiments.

## 04 / PRACTICAL NOTES

Cold-spray experiments will take place at ReliBond, with microscopy, XRD and mechanical testing at DTU. Electrical testing is outside the scope of the proposed project. The experimental plan may be refined based on initial findings and powder availability.

PROJECT LEVEL: MASTER OR BACHELOR

# Protecting steels against high-temperature ammonia corrosion




Cross-sectional BSE-SEM image showing localized substrate attack beneath a PVD coating after $NH_3$ exposure.

SUPERVISORS

**Lead Supervisor**

**Marcel A.J. Somers**
DTU Construct
majs@dtu.dk

**Co-supervisor**

**Basit Ali**
DTU Construct
baali@dtu.dk

COLLABORATION

Danish Technological Institute (DTI)

## 01 / PROJECT BACKGROUND

Ammonia is a promising carbon-free fuel and hydrogen carrier, but hot ammonia can cause nitridation and embrittlement of steels used in heat exchangers, crackers and fuel systems. Magnetron-sputtered coatings offer a possible barrier, yet local defects may allow severe attack beneath the coating. In collaboration with the Danish Technological Institute (DTI), this project investigates coating protection and failure in stainless steels.

## 02 / PROJECT AIM

Relate coating composition, thickness and defects to local ammonia attack, and identify promising protective systems under the tested conditions. Compare selected metallic and nitride coatings with uncoated references using specimens from existing and new exposure campaigns.

## 03 / PROJECT ACTIVITIES

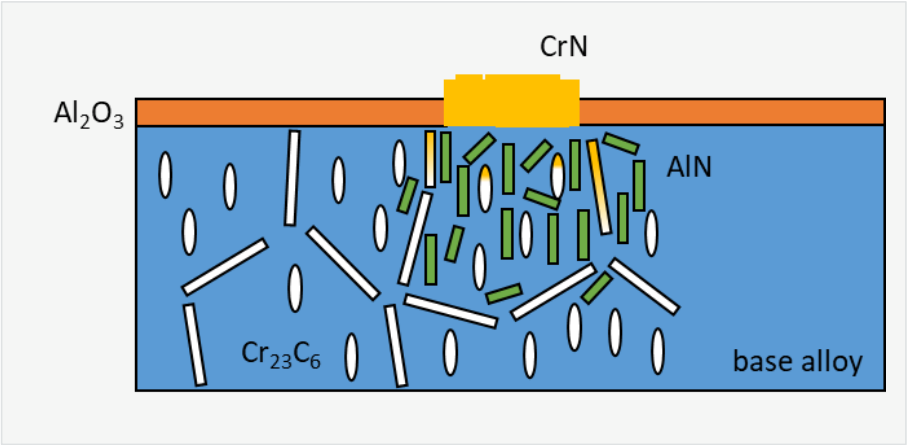
- Perform selected exposure experiments in dry and wet ammonia at controlled temperatures and gas flows
- Examine coating surfaces and polished cross-sections using light optical microscopy and SEM/EDS.
- Use XRD to identify crystalline phases and changes associated with ammonia exposure.
- Quantify attack depth, affected fraction and coating damage, then relate these to mass change and coating structure.

## 04 / PRACTICAL NOTES

DTI will supply coated specimens. Exposure experiments and characterization will take place at DTU, with training in the relevant methods provided. The scope will adapt to specimen availability, initial findings and the project duration.

PROJECT LEVEL: MASTER OR BACHELOR

# Aluminium diffusion layers for mitigation of nitridation corrosion in ammonia cracking equipment




Schematic nitridation of a cast Fe-Cr-Al alloy (from Aymans, et al. Materials & Corrosion, 2026)

SUPERVISORS

**Marcel A.J. Somers**
majs@dtu.dk

**Basit Ali**
baali@dtu.d

COLLABORATION

Materials & Corrosion-KnowHow (NL)
Chromin Maastricht (NL)

## 01 / PROJECT BACKGROUND

At elevated temperatures, ammonia dissociates at metallic surfaces and provides a high nitrogen activity at the metal surface, potentially resulting in nitridation corrosion. The project investigates the suitability of an aluminium diffusion layer against nitridation under conditions relevant to the ammonia heating section of an ammonia cracking installation.

## 02 / PROJECT AIM

The main objective of the project is to experimentally determine the effectiveness of an aluminium diffusion layer in preventing or minimizing nitridation of selected engineering materials exposed to ammonia-containing environments at approximately 250–550 °C. The investigation should provide a technical basis for assessing whether aluminium diffusion treatment is a feasible mitigation for components in the heating section of the ammonia cracking service.

## 03 / PROJECT ACTIVITIES

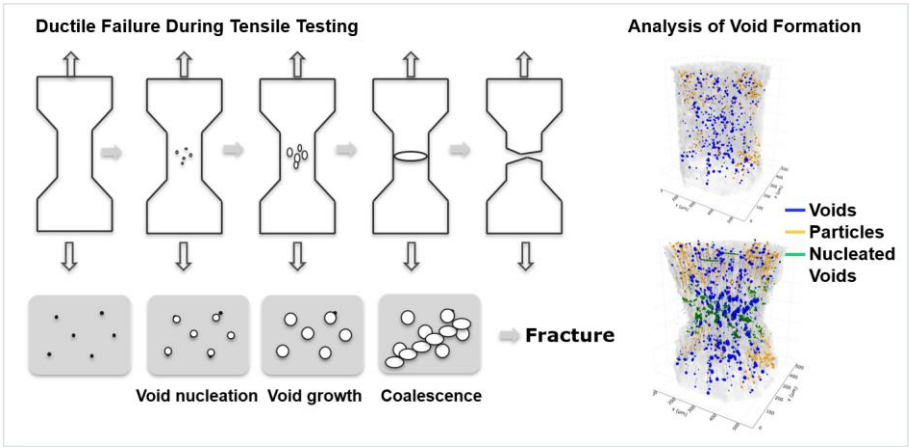
- exposure of industrially provided coated and uncoated materials to ammonia in thermobalance and stationary furnace
- characterization of the resulting microstructure with microscopy, diffraction and spectroscopy techniques
- assessment of effectictiveness of alumina coatings and potential limitations

## 04 / PRACTICAL NOTES

You will be involved in the exposure and characterization with LOM, SEM, XRD and EDS of several alloys for relatively low T nitridiation effects and determine in which temperature ranges these alloys are safe to use.

PROJECT LEVEL: Master

# X-Ray Tomography study on Void Growth in Ductile Failure




Workflow of the Project

SUPERVISORS

Felix Frankus
fefra@dtu.dk

Michela La Bella
mlabe@dtu.dk

Grethe Winther
grwi@dtu.dk

COLLABORATION

DTU 3D Imaging Centre

## 01 / PROJECT BACKGROUND

Ductile failure is often desired because the material undergoes severe plastic deformation before rupturing. It requires high ductility, allowing the material to deform plastically under load. Failure occurs as voids form within the grain structure, typically at inclusions, second-phase particles, or grain boundaries. These voids grow and coalesce, progressively reducing the load-bearing volume. X-ray tomography can track this void evolution in situ and in 3D.

## 02 / PROJECT AIM

This project aims to study the in-situ formation of voids in AA1050 and AA2219 aluminum alloys and in low-alloyed steels using X-ray computed tomography. You will conduct tensile tests in a miniature load frame inside the X-ray scanner to track void nucleation, growth, and coalescence until failure, quantifying void count, size, and location. Comparing the materials will reveal the role of second-phase particles in void formation.

## 03 / PROJECT ACTIVITIES

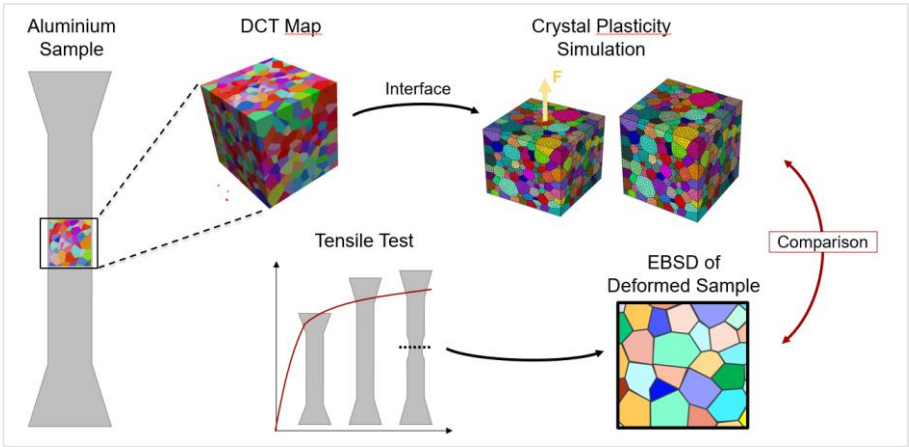
- Acquisition of X-ray CT-scans in the 3D Imaging Centre
- Tensile testing and evaluation of stress-strain curves
- Analysis of the CT-scans using particle tracking and segmentation algorithms.

## 04 / PRACTICAL NOTES

As the project deals with 3D imaging data and segmentation algorithms to track voids and second-phase particles. It will be necessary to conduct the 3D imaging analysis with Python and standard imaging analysis Libraries such as Scipy.

PROJECT LEVEL: Master

# Evolution of Grain Orientation During Tensile Deformation




Workflow of the Project

SUPERVISORS

**Felix Frankus**
fefra@dtu.dk

**Grethe Winther**
grwi@dtu.dk

**Oleg Mishin**
olmi@dtu.dk

COLLABORATION

DTU 3D Imaging Centre

## 01 / PROJECT BACKGROUND

Grain orientation strongly controls how metals deform, because it constrains each grain's response to load and its interaction with neighboring grains. Crystal Plasticity (CP) models capture these effects by simulating a meshed 3D grain map that encodes each grain's shape and orientation. Laboratory Diffraction Contrast Tomography (Lab DCT) can produce such maps non-destructively. However, few studies test whether a CP simulation seeded with a measured grain map reproduces both the macroscopic response and the grain-level deformation of the same specimen.

## 02 / PROJECT AIM

This project compares simulated tensile tests using Crystal Plasticity, seeded with a Lab DCT grain map of a polycrystalline commercially pure aluminum sample, against experimental uniaxial tensile tests on the same specimen. To connect the two, you will develop code that imports the voxelized Lab DCT data into Neper, a meshing tool for crystal plasticity data generation, and then into the Finite Element Polycrystal Plasticity program FEPX. The simulated tensile test will be compared with experimental data from the previously scanned sample. After deformation, Electron Backscatter Diffraction (EBSD) data will be used to compare grain orientations and morphology between simulation and experiment.

## 03 / PROJECT ACTIVITIES

- Acquisition of the grain maps using the LabDCT equipment in the 3D Imaging Centre
- The interfacing between the grain map from LabDCT to Crystal Plasticity
- Conducting a tensile test and characterizing the previously 3D-scanned region after deformation using EBSD

## 04 / PRACTICAL NOTES

As the project deals with complex data structures, you should be confident in using Python for analysis and interfacing between the different experimental and simulation data types and structures. Basic knowledge of Linux is beneficial for executing the simulations.

PROJECT LEVEL: Master

# Thermoresponsive Photopolymer Resins for 4D Volumetric Additive Manufacturing




Concept: VAM-printed PNIPAM soft gripper closing above its LCST

SUPERVISORS

Masoud Hasany
[mahas@dtu.dk]

Mehdi Mehrali
[meme@dtu.dk]

COLLABORATION

Aminul Islam
[mais@dtu.dk]

## 01 / PROJECT BACKGROUND

4D printing creates three-dimensional objects that change their shape or dimensions in response to an external stimulus, such as temperature. Poly(N-isopropylacrylamide) (PNIPAM) hydrogels are suitable candidates because they release water and shrink when heated above their thermal transition, and absorb water and swell again when cooled. Volumetric additive manufacturing (VAM) offers a route to printing these materials by curing selected regions within a resin volume using projected light. However, the resin must allow light to penetrate the full printing volume and gel under controlled exposure. Developing a formulation that meets these printing requirements while retaining reversible thermoresponsive behaviour after crosslinking is the central challenge of this project.

## 02 / PROJECT AIM

The project aims to develop and validate a PNIPAM-based photopolymer resin for printing thermoresponsive hydrogel structures using VAM. The formulation should combine sufficient optical transparency, suitable gelation kinetics, and a thermal transition within the 25–40 °C range after crosslinking. The project will establish suitable printing conditions and assess how accurately the printed structures reproduce their intended geometry. It will then determine whether the printed hydrogels retain reversible swelling and shrinkage over repeated heating and cooling cycles.

## 03 / PROJECT ACTIVITIES

Month 1 · **Set-up:** review PNIPAM photoresins and VAM literature; lab safety and VAM printer induction

Months 1–2 · **Formulation:** screen two PNIPAM resin strategies; tune photoinitiator and TEMPO inhibitor levels

Month 2 · **Resin characterisation:** optical clarity by UV-Vis, LCST by DSC, gel-point kinetics by photo-rheometry

Months 3–4 · **VAM printing:** calibrate light dose; print test geometries; verify print fidelity by micro-CT

Months 4–5 · **4D response:** swelling by gravimetry and DIC over 25–40 °C cycles; print a soft-gripper demonstrator

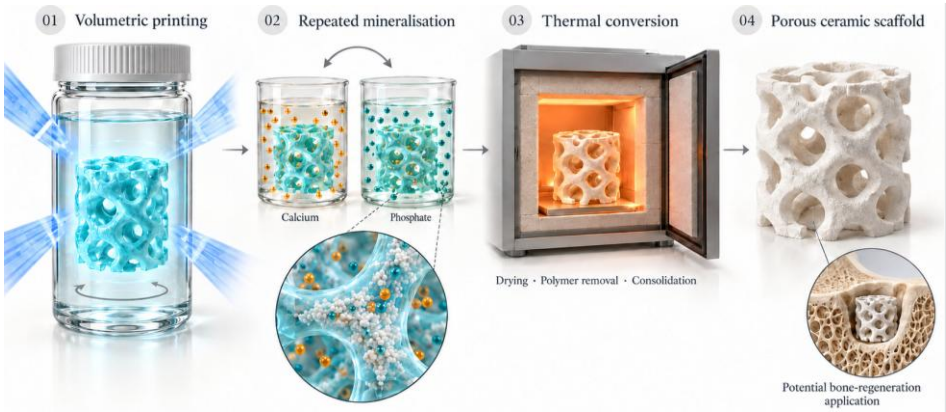
Month 5 · **Reporting:** thesis write-up, documented data set and final presentation to the group

## 04 / PRACTICAL NOTES

Experimental MSc project combining polymer formulation, additive manufacturing, and hydrogel characterization. An interest in polymer chemistry and laboratory work is required.

PROJECT LEVEL: Master

# Calcium-Phosphate Ceramic Scaffolds from Volumetrically Printed Hydrogel Templates




Proposed fabrication of porous calcium-phosphate ceramic scaffolds through volumetric hydrogel printing, repeated mineralisation, and thermal conversion.

SUPERVISORS

**Masoud Hasany**
[mahas@dtu.dk]

**Mehdi Mehrali**
[meme@dtu.dk]

COLLABORATION

**Aminul Islam**
[mais@dtu.dk]

## 01 / PROJECT BACKGROUND

Calcium-phosphate ceramics resemble the mineral component of bone and are widely investigated as scaffolds for bone regeneration. Their performance depends on both material composition and architecture, including pore connectivity, strut dimensions, and defects introduced during fabrication. Producing complex, interconnected structures while maintaining sufficient mechanical integrity remains a challenge. Volumetric additive manufacturing (VAM) enables rapid fabrication of three-dimensional structures within a photosensitive resin. However, incorporating ceramic particles directly into the resin can cause light scattering, restricting the achievable inorganic loading and printing accuracy. An alternative is to print a transparent hydrogel template first and introduce the inorganic material afterward.

## 02 / PROJECT AIM

This project explores repeated calcium-phosphate mineralization of VAM-printed hydrogel templates, followed by polymer removal and thermal consolidation. The central challenge is to form a sufficiently continuous mineral network throughout the template so that the resulting ceramic retains its architecture. The work will investigate how hydrogel composition and mineralization conditions influence this conversion. The intended outcome is a processing–structure–property relationship that identifies conditions suitable for producing ceramic scaffolds with preserved interconnected pores and reproducible mechanical properties. The materials will be developed as candidates for future bone-regeneration applications.

## 03 / PROJECT ACTIVITIES

Month 1 · Review hydrogel mineralisation, calcium-phosphate ceramics and VAM literature; lab safety and printer
Months 1–2 · Prepare hydrogel specimens, screen repeated calcium-loading and phosphate-exposure cycles
Months 3–4 · VAM-print porous lattices, refine mineralisation
Months 4–5 · Measure dimensional shrinkage and compression properties; assess pore connectivity and defects by micro-CT and microscopy
Month 5 · Thesis write-up, documented data set and final presentation to the group

## 04 / PRACTICAL NOTES

Experimental MSc project combining polymer formulation, additive manufacturing, and biomedical engineering. An interest in polymer chemistry and laboratory work is required.